

When induced gravity attracts: an exact count for the Standard Model

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In induced gravity, Newton’s constant arises from the one-loop vacuum energy of matter, and its sign is the sign of a sum over species, $N_{\text{net}} = \sum \sigma(N - 6c)$, with σ the statistics sign, N the number of field components and cR the trace of the curvature term in the species’ own fluctuation operator. Under a spectral cutoff taken in each species’ own operator, N_{net} is +1 for the Standard Model with the Higgs minimally coupled, in units of one minimally coupled scalar, the value an earlier species count also reaches. Its kinetic parts σN sum to -62 and its curvature insertions $-6\sigma c$ to $+63$. The cutoff weights each insertion at exactly six times its kinetic part, for every cutoff function of bounded variation, so the count is exact and does not depend on the cutoff function; the sign would reverse only at a weighting of $124/21$, 1.59% below six, which only a cutoff departing from this condition could reach. The chain from the regulated action to the sign is proved for proper-time cutoffs, and for other cutoff functions rests on the spectral-action expansion as the literature states it. A horizon-entropy count, entanglement $+73$ against the gauge fields’ contact term -72 , agrees with it species by species, a consistency check on the species table. The count is an integer and $+1$ is its least positive value; with the Standard Model’s gauge group, three generations and one Higgs doublet are the only content of whole generations and doublets that reaches it. Attraction begins at three generations, a floor rather than a selection. The count requires the Higgs doublet’s common curvature coupling to stay below $1/24$, and keeps its sign for every number of right-handed neutrinos. If Newton’s constant is entirely induced, as Sakharov proposed, the count becomes a condition on content, much as anomaly cancellation is: each gauge boson below the cutoff must be paid for by four Weyl fermions or minimally coupled real scalars, and a conformally coupled Higgs doublet is excluded, while Higgs inflation, whose coupling has the other sign, is not. The counts and the identities they rest on are machine-checked in Lean.

I. INTRODUCTION

Sakharov proposed that the Einstein action is not fundamental: it is the change in the one-loop action of quantum fields when spacetime is curved [1]. Amati and Veneziano extended the idea to the gauge couplings, inducing them from the same matter content [2], and Visser reviews the mechanism in modern form [3]. In each version the coefficient of $\int \sqrt{g} R$ in the induced action is a sum over species, each weighted by the first heat-kernel coefficient of its own operator [4, 5].

That makes the *sign* of the induced Newton constant the sign of a sum over species, the net count N_{net} of Eq. (5). Each species contributes a kinetic part, from the Laplacian acting on its field components, and a curvature insertion, from the curvature term in its own operator. For the fermions and the gauge fields the two have opposite signs (for minimally coupled scalars and for ghosts the insertion vanishes), so whether the matter we have induces attractive gravity depends on what that matter is. In units of one minimally coupled scalar, a scalar counts $+1$, a Weyl fermion $+1$, and a gauge field with its two ghosts -4 . The Standard Model’s 4 scalars and 45 Weyl fermions give $+49$, its 12 gauge fields -48 , and the net is $+1$. Figure 1 shows the count for the Standard Model’s gauge group, one Higgs doublet, and any number of generations.

The count has been made before. Tanaka computed the induced Einstein action in supersymmetric models, found that vector multiplets induce a negative Newton constant and scalar multiplets a positive one, and concluded that the minimal supersymmetric Standard Model

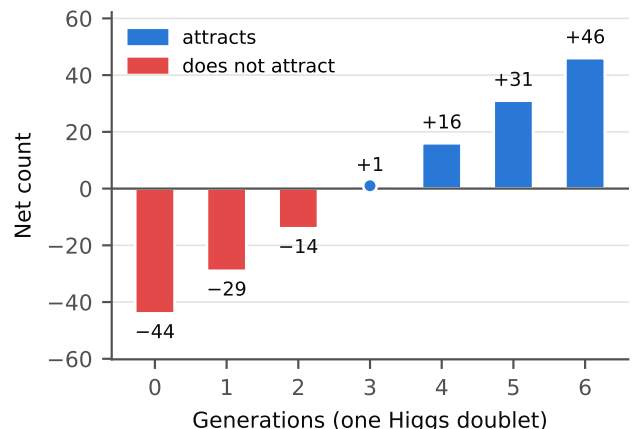


FIG. 1. The net count N_{net} of Eq. (5), in units of one minimally coupled scalar, against the number of generations, with the Standard Model’s gauge group and one Higgs doublet. Gravity attracts exactly when $N_{\text{net}} > 0$. The count is negative up to two generations and positive from three on, where it is $+1$ (marked, since the bar is too short to see).

and minimal supersymmetric $SU(5)$ need three generations or more for attractive gravity [6]. Broda and Szanecki weighted the species exactly as here, under a Planckian proper-time cutoff [7]. For the Standard Model with its Higgs doublet and twelve gauge fields, their combination of species equals one, and they note that the black hole’s geometric entropy carries the same combination. They call the gauge fields’ contribution disputable,

since the gauge fields are themselves induced. The net count of this paper is therefore not new.

What this paper adds is the count's structure, and its conditions. Resolved into its parts, the kinetic parts sum to -62 and the curvature insertions to $+63$. The weighting that the cutoff gives the insertions is exactly six for every profile of bounded variation, by an identity between the profile's moments, so the count is exact and the profile cannot change it; for proper-time profiles, the heat kernel's among them, the whole chain from the regulated action to the sign is proved, the one-to-six ratio derived rather than quoted (Sec. III, Appendices A–C). An exact criterion locates the boundary: the sign reverses at a weighting of $124/21$, 1.59% below six, which only a cutoff departing from that condition could reach (Sec. IV). A horizon-entropy count agrees with the count species by species, a consistency check on the species table and on condition K2 (Sec. V). The count is an integer, so $+1$ is the least positive value it can take, and with the gauge group given, the Standard Model's generations and doublet are the only content of whole generations and Higgs doublets that reaches it. The same count gives a floor on the number of generations, a bound on the Higgs doublet's curvature coupling, and the effect of right-handed neutrinos (Sec. VI). Under Sakharov's premise that Newton's constant is entirely induced, it becomes a condition on content: gauge bosons must be paid for in matter, and a conformally coupled Higgs doublet is excluded (Sec. VIII). Every count, and the identities behind them, is machine-checked in Lean and cross-checked against a second implementation, both reading one table of species with a citation for each entry (Sec. X).

The paper does not claim the magnitude of Newton's constant, which moves with the cutoff's scale and the profile's shape and is reported only as a range (Sec. VII). Nor does it claim anything about an induced graviton's dynamics, or about the vacuum energy (Sec. IX).

II. THE CONDITIONS

The count is made under four conditions, stated in Table I. Of the four, K1 is the one the sign depends on; K2 decides whether the count is $+49$ or $+1$; K3 and K4 limit what is claimed.

Under K1, with the cutoff taken in each species' own fluctuation operator, the curvature term of that operator enters the heat kernel's first coefficient on the same footing as the Laplacian, and the insertion carries the same moment of the cutoff function as the kinetic part (Sec. III). A cutoff placed in another operator can reweight the insertion, and the criterion of Sec. IV then decides whether the sign survives.

Condition K2 takes the gauge fields as fundamental fields up to the cutoff, each with its two ghosts and its contact term with the horizon. It is the condition Broda and Szanecki question, since the gauge fields may themselves be induced [7]. Without the gauge fields the Stan-

dard Model's count would be $+49$; counting them with their ghosts adds -48 and brings it to $+1$. K2 therefore leaves the sign positive and brings the count to its least positive value.

The regulator here is covariant. Condition K3 records that what is computed is the coefficient of $\int \sqrt{g} R$ in the Euclidean action on a fixed background, not an induced graviton's propagation, whose time and space derivative terms a non-covariant regulator could treat differently [3]. Condition K4 sets the cosmological term aside.

a. Further assumptions. Besides K1 to K4, the count assumes: free massless fields on a purely gravitational background; a closed Euclidean background with $\int \sqrt{g} R \neq 0$, which makes the induced coefficient unique; a cutoff function whose moment M_f (Eq. (3)) is positive, which holds for every proper-time cutoff and every non-increasing one; each gauge vector in Feynman gauge with its two Faddeev–Popov ghosts, counted as an abelian field, which is exact at one loop on a purely gravitational background, with the dependence on the gauge parameter not examined; and, for the main result, a minimally coupled Higgs. Appendix A states each in the form the proofs use.

III. EACH SPECIES' WEIGHT

A *species* is a field with its own fluctuation operator; the Faddeev–Popov ghosts count as species of their own, and a *content* is a list of species with their multiplicities. Each species' fluctuation operator is of Laplace type, $D = D_0 + X$ with $D_0 = -\nabla^2$ acting on N field components, and a curvature term X whose trace is $\text{tr } X = cR$. Its heat trace has the expansion [5]

$$\text{Tr } e^{-\tau D} = \tau^{-d/2} [a_0 + a_1 \tau + O(\tau^2)], \quad (1)$$

whose curvature coefficient is

$$a_1 = (4\pi)^{-d/2} \left(\frac{N}{6} - c \right) \int \sqrt{g} R. \quad (2)$$

The Euclidean one-loop action of a content is $W = \frac{1}{2} \sum_i \sigma_i \log \det D_i$, with σ_i the statistics sign. The cutoff of K1 regulates the logarithm species by species: $\log \det D$ is replaced by $-\text{Tr } f(D/\Lambda^2)$, for a cutoff function f (the *profile*), such as the sharp step $\theta(1-u)$, the heat kernel's own e^{-u} , or a Gaussian e^{-u^2} . So $W_f(\Lambda) = -\frac{1}{2} \sum_i \sigma_i \text{Tr } f(D_i/\Lambda^2)$. At order Λ^{d-2} each trace carries the heat kernel's a_1 times one moment of the profile [8–11],

$$M_f = \frac{1}{\Gamma(s)} \int_0^\infty f(u) u^{s-1} du, \quad s = \frac{d}{2} - 1, \quad (3)$$

normalized to the heat kernel's own profile, for which $M_{e^{-u}} = 1$. In the Euclidean convention $W \supset -(16\pi G)^{-1} \int \sqrt{g} R$ [12], the induced coupling is then

$$\frac{1}{16\pi G} = \frac{M_f}{12(4\pi)^{d/2}} N_{\text{net}} \Lambda^{d-2}, \quad (4)$$

TABLE I. The conditions the count is made under.

Condition	Why it matters
K1 The cutoff is spectral, taken in the fluctuation operator, with its profile normalized to the heat kernel's	Under this cutoff the kinetic part and the curvature insertion stand at one to six for every profile. A cutoff taken in another operator transfers a share of that operator's curvature, and the ratio is lost
K2 One loop, with the gauge fields counted as species at the cutoff, each with its contact term	The contact term is what reconciles the induced coupling with horizon entropy
K3 The static coefficient only. The paper claims Newton's constant, not an induced graviton sector	With a regulator that is not Lorentz-invariant, an induced action's time and space derivative terms need not agree, so a dynamical claim would need more than a static one
K4 The vacuum energy is not addressed. The cosmological term is set aside, and the count concerns the curvature term alone	The cosmological term is the cutoff trace's leading order and the curvature term the next, so the count neither needs the vacuum energy nor says anything about it

with the content's net count

$$N_{\text{net}} = \sum_i \sigma_i (N_i - 6c_i), \quad (5)$$

so the sign of G is the sign of N_{net} for every profile with a positive moment, and gravity attracts exactly when $N_{\text{net}} > 0$.

Each species' weight $\sigma(N - 6c)$ is the sum of a *kinetic part* σN , from D_0 , and a *curvature insertion* $-6\sigma c$, from X . Table II gives the four species the count needs. The weights reproduce the cited values: +1 for a minimal scalar, +1 for a Weyl fermion, and -4 for a gauge field with its two ghosts [7, 13]. The conventions are in Appendix D.

The claim that the profile cannot change the sign rests on two steps, with different scopes.

a. Step 1: the moments. The insertion can be read in two ways. In Eq. (2) it is part of the full operator's coefficient, and carries the profile's moment M_f . Read instead as the first-order response of the cutoff trace to the curvature term, $\frac{d}{d\varepsilon} \text{Tr} f((D_0 + \varepsilon X)/\Lambda^2)$ at $\varepsilon = 0$, as a perturbative treatment of X would compute it, it carries the profile's *derivative* moment, $-\int_0^\infty f'(u) u^s du$, divided by $\Gamma(s+1)$. The two agree because of an integration by parts,

$$-\int_0^\infty u^s df(u) = s \int_0^\infty f(u) u^{s-1} du, \quad (6)$$

which holds for every profile whose derivative is a measure: every profile of bounded variation with finite moments, the sharp step included. For the sharp step, $f' = -\delta(u-1)$, the left side is one and the right side is $s \cdot (1/s)$. So the insertion's weight relative to the kinetic part is six for every profile of bounded variation (Appendix C). This step is an identity between moments; it does not yet say that the moments are what the cutoff trace carries.

b. Step 2: the expansion. That the cutoff trace carries $M_f a_1$ at order Λ^{d-2} , so that the profile enters Eq. (4) as one positive factor, is the large- Λ expansion

of $\text{Tr} f(D/\Lambda^2)$ in heat-kernel coefficients. Its standing depends on the profile:

- For *proper-time* profiles, $f(u) = \int e^{-tu} d\nu(t)$ with ν a finite measure on a bounded proper-time interval, the heat kernel's own profile among them, the expansion is a theorem: it follows from the heat trace alone (Appendix B). For them the two readings of the insertion are both computed and shown to agree, so the one-to-six ratio is derived, not quoted (Appendix C), for an insertion that is constant over the background.
- For other *Laplace-transform* profiles and for *Schwartz* profiles the expansion is stated in the spectral-action literature [9–11]. It is taken as a hypothesis for them, not proved. The spectral action of Ref. [8] likewise puts its cutoff in an operator, the Dirac operator, though there the cutoff trace defines the bosonic action rather than regulating each species' one-loop trace.
- For the *sharp step* the identity (6) holds exactly, but its cutoff trace has the expansion beyond the leading term only as a Cesàro mean [11]: the subleading Λ^{d-2} term exists only after averaging over the cutoff scale. The sharp step is therefore covered by Step 1, and by Step 2 only in that averaged sense.

IV. THE STANDARD MODEL'S COUNT

a. The count. The Standard Model with one Higgs doublet carries 4 real scalars, 45 Weyl fermions in 3 generations, 12 gauge vectors and their 24 ghosts [14]. Table III gives the count species by species. The kinetic parts sum to $N_{\text{kin}} = -62$ and the insertions to $N_{\text{ins}} = +63$, a net of +1: with the Higgs minimally coupled, the Standard Model induces a positive Newton constant under K1.

TABLE II. Each species' statistics sign σ , field components N , curvature trace c and weight, with its coefficient in a horizon's entanglement entropy and its contact term with the horizon (Sec. V), both in units of one minimally coupled real scalar. The entanglement coefficient is not a count of physical degrees of freedom: a Weyl fermion has two, which count half each, as fermions do. The contact term is Kabat's [13].

Species	σ	N	c	Weight	$\sigma(N - 6c)$	Entanglement	Contact
minimal scalar	+1	1	0		+1	1	0
Weyl fermion	-1	2	1/2		+1	1	0
gauge vector	+1	4	1		-2	2	-6
ghost	-1	1	0		-1	0	0

TABLE III. The Standard Model's count, species by species, in units of one minimally coupled scalar: each species' kinetic part is $m\sigma N$ and its insertion $-6m\sigma c$, for multiplicity m . The last two columns are the horizon count's (Sec. V).

Species	Multiplicity	Kinetic	Insertion	Weight	Entanglement	Contact
minimal scalar	4	+4	0	+4	+4	0
Weyl fermion	45	-90	+135	+45	+45	0
gauge vector	12	+48	-72	-24	+24	-72
ghost	24	-24	0	-24	0	0
Total		-62	+63	+1	+73	-72

b. The criterion. Weight every species' insertion by ρ in place of six. The count $N_{\text{net}}(\rho) = N_{\text{kin}} + (\rho/6)N_{\text{ins}}$ is positive exactly when

$$\rho > \frac{6|N_{\text{kin}}|}{N_{\text{ins}}} = 124/21 \approx 5.905. \quad (7)$$

Condition K1 gives $\rho = 6$ for every profile of bounded variation, so under K1 ρ is not a free parameter and the count is exactly +1. The criterion locates the boundary instead. The *margin*, the threshold's distance below six as a fraction of six, is $(6 - 124/21)/6 = 1/63$, or 1.59%; it equals $N_{\text{net}}/N_{\text{ins}}$, the net count over the insertions' total, and is 1/23 and 1/18 with two and three right-handed neutrinos (Table IV). It is how far a scheme departing from K1 would have to lower the insertions' common weight to reverse the sign; no such scheme is analysed here. The criterion takes one ρ for every species: a scheme that reweighted fermion and vector insertions differently, whose contributions have opposite signs, would need the criterion species by species. Figure 2 shows the count against ρ for the Standard Model with no, two and three right-handed neutrinos (Sec. VI).

V. THE HORIZON ROUTE

Larsen and Wilczek showed that the one-loop correction to a black hole's entropy and the renormalization of Newton's constant carry the same divergence [15]. In induced gravity, the horizon entropy is the fields' entanglement entropy less contact terms with the horizon [12], and once G is the induced constant, the Bekenstein-Hawking entropy $A/4G$ requires the two counts to agree. Each species enters the entanglement entropy with a coefficient that, in units of one real scalar, is 1 for a scalar,

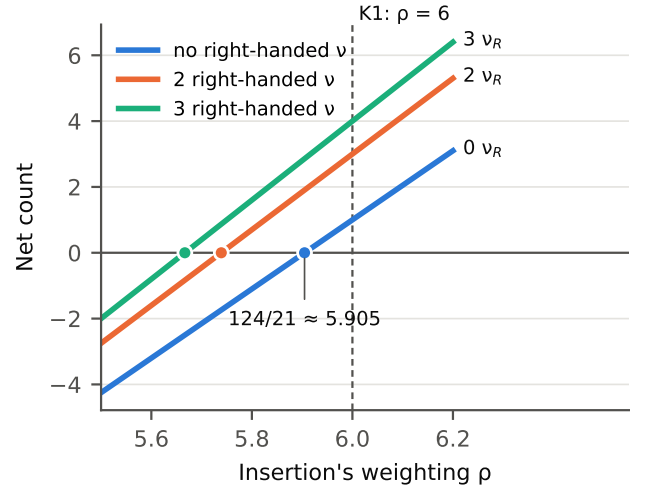


FIG. 2. The net count against ρ , the weight of every species' curvature insertion relative to its kinetic part (Eq. (7)), for the Standard Model with no, two and three right-handed neutrinos. Each line crosses zero at its content's threshold (dots): 124/21, 132/23 and 17/3 respectively. The cutoff of K1 gives $\rho = 6$ (dashed).

1 for a Weyl fermion (whose two physical degrees of freedom count half each, as fermions do), 2 for a gauge field's two polarizations, and 0 for a ghost (Table II). A gauge field adds a contact term with the horizon, -6 in units of one minimal scalar [13], which Donnelly and Wall read as the entanglement of edge modes [16]; the count uses only its coefficient, which the two readings share. Solodukhin reviews both relations [17].

The last two columns of Table III give the Standard Model's horizon count: entanglement +73 and contact

term -72 , a net of $+1$, as the proportionality requires. The agreement holds species by species. A gauge field costs four by either route: in the induced count its weight -2 and its two ghosts' -1 each, and in the horizon count its two polarizations and its contact term -6 . The agreement therefore holds for every content whose vectors each carry their two ghosts. It is a consistency check on the species table and on condition K2, not independent evidence for the sign of the Standard Model's count. Broda and Szanecki note the same agreement for the geometric entropy, their name for the entanglement entropy [7].

VI. COROLLARIES

A content with the Standard Model's gauge group, g generations, h Higgs doublets and n right-handed neutrinos has the count

$$N_{\text{net}} = 15g + 4h + n - 48, \quad (8)$$

the gauge sector contributing -48 . The floor, the least content and the right-handed neutrinos' effect hold for every g , h and n ; the coupling bound is stated for the Standard Model, and in general form in Appendix D.

a. A floor on generations. With one doublet, the count runs -44 , -29 , -14 and $+1$ for zero to three generations (Fig. 1), and it is positive for every number of generations from three on. Attraction begins at three generations. That is a floor, not a selection: every larger number also attracts. Tanaka found the same floor for supersymmetric contents [6]; here it holds for the Standard Model's own content, with the gauge fields counted, and three generations clear it by one.

b. The least content. The count is an integer, so $+1$ is the least positive value it can take. Among contents of whole generations and Higgs doublets with this gauge group, the only one that reaches it is three generations and one doublet. By Eq. (8) a count of one needs $15g + 4h = 49$, which allows at most three generations, and of those only three with one doublet solves it. With a right-handed neutrino in every generation, every term of the count is a multiple of four, so no such content has a count of one. Both statements hold for every number of generations and doublets; Appendix D gives the finite search that cross-checks them. Other representations are outside the statement: adding singlet scalars, for instance, would give other contents of count one.

c. The Higgs coupling. Give the doublet's 4 real scalars a common coupling ξ to curvature, with $\xi = 0$ minimal and $\xi = 1/6$ conformal [4]. Each scalar's curvature trace becomes ξ , and the count falls by 6ξ per scalar. Attraction then requires

$$\xi < 1/24, \quad (9)$$

strictly, since the count vanishes at the bound. With two and three right-handed neutrinos the bound rises to $1/8$ and $1/6$. A coupling below the bound, of either sign,

TABLE IV. The Standard Model with ν_R right-handed neutrinos added: the count, the threshold weighting, the margin, and the bound on the Higgs doublet's common curvature coupling.

ν_R	Count	Threshold ρ	Margin	Bound on ξ
0	+1	124/21	1.59%	1/24
2	+3	132/23	4.35%	1/8
3	+4	17/3	5.56%	1/6

TABLE V. The profile's moment M_f and the cutoff in Planck units, by profile and number of right-handed neutrinos ν_R . The heat kernel's row rests on the proved expansion, the Gaussian's on the imported one for Schwartz profiles; the sharp step's moment is exact, but its expansion beyond the leading term holds only as a Cesàro mean (Appendix E).

Profile	M_f	Λ/M_P		
		$\nu_R = 0$	$\nu_R = 2$	$\nu_R = 3$
sharp step	1	6.140	3.545	3.070
heat kernel e^{-u}	1	6.140	3.545	3.070
Gaussian e^{-u^2}	0.8862	6.522	3.766	3.261

keeps the count positive; what the bound excludes, if gravity is induced, is taken up in Sec. VIII.

d. Right-handed neutrinos. Each right-handed neutrino adds one to the count. With two the count is $+3$, the threshold $132/23$ and the margin 4.35% ; with three, $+4$, $17/3$ and 5.56% (Table IV). The rows are none, the Standard Model's own content; two, the least number that gives the two non-zero neutrino masses the observed oscillations need [14], as in the minimal seesaw of Framp-ton, Glashow and Yanagida [18]; and three, one per generation. Eq. (8) gives the count for any number, and the sign holds for every one.

VII. THE MAGNITUDE, AND WHY IT IS NOT CLAIMED

In four dimensions Eq. (4) gives the cutoff at which a content induces Newton's constant,

$$\Lambda = M_P \sqrt{\frac{12\pi}{M_f N_{\text{net}}}}, \quad M_P = G^{-1/2}. \quad (10)$$

The profile's moment is one for the sharp step and for the heat kernel, and 0.8862 for the Gaussian e^{-u^2} (Appendix E). For the Standard Model the required cutoff ranges from 6.140 to $6.522 M_P$ over these profiles (Table V), and right-handed neutrinos lower it. That a net of one needs a cutoff a few times the Planck mass is the same statement as Broda and Szanecki's coefficient $1/12\pi$ at a Planckian cutoff [7]. The paper does not predict the cutoff's scale or its profile; given them, this equation fixes the magnitude. The magnitude moves with the profile. The sign does not.

VIII. WHAT THE COUNT FORBIDS, IF GRAVITY IS INDUCED

Everything so far concerns the sign of the curvature term a content induces. It becomes a constraint on content under one further premise: that Newton’s constant is *entirely* induced, with no bare term, by the species lighter than the cutoff. That is Sakharov’s premise [1], and this paper does not argue for it. Without it, a bare term can carry the sign, and the count gives only the sign of matter’s contribution to $1/G$; nothing below is then forbidden. With it, and under K1 to K4, a content whose count is not positive induces no attractive gravity. The count then acts as a condition on what the content can be, much as anomaly cancellation does: it excludes contents without predicting a number. Unlike anomaly cancellation, it holds only under the premise.

a. Gauge bosons must be paid for. A gauge boson with its two ghosts contributes -4 by either route (Sec. V), and a Weyl fermion or a minimally coupled real scalar contributes $+1$. A content of these species therefore has $N_{\text{net}} = n_{\text{scalar}} + n_{\text{Weyl}} - 4n_{\text{vector}}$, the combination of Broda and Szanecki [7], and the Standard Model, with a count of $+1$, has one unit to spare. Hence:

- the Standard Model with one further gauge boson below the cutoff, and nothing else, has count -3 ;
- with a complex scalar added, minimally coupled, to give that boson its mass, the count is -1 ;
- the Standard Model extended by b gauge bosons and k further Weyl fermions or minimally coupled real scalars has a count $k - 4b$ above the Standard Model’s, and attracts exactly when $k \geq 4b$ (Appendix D).

Each new gauge boson must bring four such fields with it. Under the premise this bears on every extension with a new gauge boson, such as a dark photon or a Z' : its content below the cutoff must pay for its bosons. A boson’s mass does not change what it costs. A mass μ shifts each species’ operator by the constant μ^2 , which changes a_1 only by $-\mu^2 a_0$ (Appendix C), a multiple of the volume rather than of $\int \sqrt{g} R$; so at order Λ^{d-2} a species lighter than the cutoff counts as a massless one does. The formal statements are for massless fields (Sec. IX) and do not include this step.

b. The Higgs doublet’s curvature coupling. Under the premise the doublet’s common coupling must satisfy $\xi < 1/24$ (Sec. VI). The conformal coupling lies above the bound: with the doublet conformally coupled, $\xi = 1/6$, the Standard Model’s count is -3 , so a conformally coupled Higgs doublet is excluded. Right-handed neutrinos leave this in place: with two the bound is $1/8$, and with three it is $1/6$, the conformal value itself, where the count vanishes. Higgs inflation is not excluded. It needs a large coupling, $\xi \approx 49000\sqrt{\lambda}$ in the convention of Ref. [19], which places the conformal coupling at $-1/6$. In this paper’s convention the same coupling is large and negative,

and it raises the count, by 24 per unit of $|\xi|$, rather than lowering it. The bound excludes large couplings of the conformal sign, not Higgs inflation’s.

c. What is not forbidden. Further generations, right-handed neutrinos and minimally coupled Higgs doublets all raise the count, so it allows them. The floor of three generations excludes only what observation already excludes.

d. How the constraints meet observation. Both constraints concern the complete content below the cutoff, which lies at a few Planck masses (Sec. VII). The matter that pays for a gauge boson may lie anywhere below it, so no single discovery can violate the gauge-boson rule: a new gauge boson found without the matter to pay for it would only place that matter higher. The rule constrains complete models, which must list their content up to the cutoff. The coupling bound is likewise a constraint on models rather than a prediction for a measurement: collider data bound the Higgs boson’s curvature coupling only at the order of 10^{15} [20], far above $1/24$.

IX. WHAT THE COUNT DOES NOT SETTLE

a. Whether gravity is induced. The constraints of Sec. VIII rest on Sakharov’s premise that Newton’s constant is entirely induced, which the paper states and does not argue for.

b. The dynamical sector. The count is the coefficient of $\int \sqrt{g} R$ on a fixed Euclidean background, not an induced graviton’s propagation; with a regulator that is not Lorentz-invariant the two need not agree [3] (K3).

c. The vacuum energy, the cutoff trace’s leading term, is set aside (K4).

d. Beyond one loop. The count is the one-loop coefficient of massless fields on a purely gravitational background. Interactions, masses and higher loops are not included.

e. Other cutoffs. A cutoff taken in an operator other than the species’ own fluctuation operator falls outside K1. Such a cutoff can move part of the curvature term out of the regulated operator, so the insertion’s weight need no longer be six, and the count is not made for it here. The sharp step is covered only in the Cesàro sense noted in Sec. III.

f. The content. The main result takes the Higgs minimally coupled, and the corollaries range over whole generations, Higgs doublets and right-handed neutrinos, not over other representations.

X. MACHINE-CHECKED STATEMENTS

The counts and the identities above are stated and proved in Lean 4 with Mathlib. The Lean proofs use a species table generated from a data file; a second implementation, in Python, reads the same file, and the build fails if the two ever disagree on a count. The proofs

contain no unproved steps (Lean’s `sorry`) and no unverified evaluation (`native.decide`), and use no axioms beyond Lean’s standard three: propositional extensionality, choice and quotient soundness.

What is proved: the weight’s decomposition into kinetic part and insertion; the integration by parts (6) for every profile of bounded variation; the expansion of the cutoff trace for proper-time profiles, from the heat trace alone; the one-to-six identity, with the insertion’s first-order response computed; the induced coupling (4) of any content; the Standard Model’s and the horizon counts; the criterion; the floor and the least content, for every number of generations and doublets; the coupling bound, for every real ξ ; the effect of right-handed neutrinos, for any number of them; the magnitude formula with the three profiles’ moments; and the cost of a gauge boson by both routes, the count of every content of scalars, Weyl fermions and gauge vectors, and the Standard Model’s extensions and its conformally coupled Higgs doublet, with the Newton constant each induces.

What is imported, as hypotheses the theorems take rather than as axioms: the heat trace’s expansion and its first coefficients [5]; the cutoff trace’s expansion for profiles outside the proper-time class [9–11]; and the species table’s entries, including the contact term [13]. The sign convention for the induced coupling is a definition [12].

a. What the checks do not reach. A proof shows that a statement holds; it does not show that the statement says what the result citing it says, that the imported inputs and the species table’s entries are the right physics, or that the conditions are the right ones. Table VI in Appendix F sets each result beside the section it appears in and the statements it cites. The two implementations, written by one author from one table, guard against slips of arithmetic rather than against a shared misreading.

XI. CONCLUSION

Gravity’s attractive sign can be read off the matter we have. Under a spectral cutoff in each species’ own operator, the Standard Model’s count is exactly +1, the least positive value the count can take. Its kinetic parts, −62, and its insertions, +63, stand at one to six for every profile of bounded variation, so the profile cannot change the count (the chain to the sign is proved for proper-time profiles, and rests on the stated expansion for others); the sign would reverse only under a cutoff that lowered the insertions’ weight by 1.59%. The count agrees species by species with the horizon count and keeps its sign for every number of right-handed neutrinos. Attraction begins at three generations, a floor rather than a selection, and requires the Higgs doublet’s curvature coupling to stay below 1/24: any model that induces gravity from this content under K1 to K4 must meet both. If Newton’s constant is entirely induced, the count becomes a condition on content: each gauge boson must be paid for by four Weyl fermions or minimally coupled real scalars,

and a conformally coupled Higgs doublet is excluded.

a. Data and code availability. Every number, table and figure in this paper is generated by the `induced-gravity-count` repository [21], where one command reproduces them from a clean checkout and the Lean proofs build with the paper’s numbers. Its `CLAIMS.md` maps each result to the computation and the Lean statements behind it, and its `LIMITS.md` states what is not established.

Appendix A: Setup: the regulated action and the Newton constant it induces

a. Spectral data. Each species’ fluctuation operator D enters only through its spectral measure μ , which counts eigenvalues with multiplicity. Its heat trace, and its cutoff trace under a profile f at scale $L = \Lambda^2$, are

$$\begin{aligned}\Theta(\tau) &= \int e^{-\tau\lambda} d\mu(\lambda), \\ \text{Tr } f(D/L) &= \int f(\lambda/L) d\mu(\lambda).\end{aligned}\tag{A1}$$

For every species, $\Theta(\tau)$ is assumed to exist for all $\tau > 0$ and, with $s = d/2 - 1$, to satisfy

$$|\tau^{s+1}\Theta(\tau) - a_0 - a_1\tau| \leq C\tau^2 \quad \text{for } 0 < \tau \leq \tau_0, \tag{A2}$$

the small- τ expansion of Ref. [5]. In four dimensions $s = 1$.

b. The background. The background is closed, with $\int \sqrt{g} R \neq 0$, and a species of N components whose operator carries the curvature term cR has $a_1 = (4\pi)^{-(s+1)}(N/6 - c) \int \sqrt{g} R$ [4, 5]. The non-vanishing integral makes the induced coefficient below unique.

c. The coefficient at order L^s (condition K4). A function $T(L)$ has coefficient b at order L^s when, for some A ,

$$L^{-s}(T(L) - AL^{s+1}) \rightarrow b \quad (L \rightarrow \infty). \tag{A3}$$

The leading term AL^{s+1} is the cosmological one. It is removed by the definition and never claimed. The coefficient, when it exists, is unique, and it is linear in T .

d. The action (condition K1) and the Newton constant. Each species is cut in its own operator:

$$W(L) = -\frac{1}{2} \sum_{\text{species}} m \sigma \text{Tr } f(D/L), \tag{A4}$$

with multiplicity m and statistics sign σ . With the Euclidean convention $W \supset -(16\pi G)^{-1} \int \sqrt{g} R$ [12], the content induces κ when W has coefficient $-\kappa \int \sqrt{g} R$ at order L^s . Then $(16\pi G)^{-1} = \kappa L^s$, and gravity attracts, $G > 0$, at every scale exactly when $\kappa > 0$.

e. The theorem. Let $M_f = \Gamma(s)^{-1} \int_0^\infty f(u) u^{s-1} du$ be the profile's moment, Eq. (3), normalized to the heat kernel's, so that $M_f = 1$ for $f(u) = e^{-u}$. If each species' cutoff trace has coefficient $M_f a_1$ at order L^s (proved for proper-time profiles in Appendix B), then

$$\kappa = \frac{M_f (4\pi)^{-(s+1)}}{12} \sum_{\text{species}} m \sigma (N - 6c), \quad (\text{A5})$$

by the linearity of (A3). The sum is the net count. For $M_f > 0$, $\kappa > 0$ exactly when the net count is positive: the profile scales the coefficient and cannot change its sign. $M_f > 0$ holds for every proper-time profile, and for every non-increasing profile with finite moments that vanishes at infinity and is not identically zero.

Appendix B: The expansion, proved for proper-time profiles

A proper-time profile is a Laplace transform,

$$f(u) = \int e^{-tu} d\nu(t), \quad (\text{B1})$$

of a non-zero finite measure ν carried by $(0, T]$ with $\int t^{-(s+1)} d\nu < \infty$. It is the regulated logarithm of every proper-time cutoff with an ultraviolet and an infrared end, and it contains the heat kernel's profile, $\nu = \delta_1$.

a. Claim. For such a profile, $\text{Tr } f(D/L)$ has coefficient $M_f a_1$ at order L^s . Nothing is assumed beyond the heat-trace bound (A2).

b. Proof. By Fubini, $\text{Tr } f(D/L) = \int \Theta(t/L) d\nu(t)$: the cutoff trace is the heat trace averaged over proper time. For $L \geq T/\tau_0$, every t in the support has $t/L \leq \tau_0$, and (A2) at $\tau = t/L$, multiplied by $L t^{-(s+1)}$, gives

$$\left| L^{-s} \Theta(t/L) - a_0 L t^{-(s+1)} - a_1 t^{-s} \right| \leq \frac{|C| t^{1-s}}{L}. \quad (\text{B2})$$

Integrating against ν , with $A = a_0 \int t^{-(s+1)} d\nu$,

$$\begin{aligned} \left| L^{-s} (\text{Tr } f(D/L) - A L^{s+1}) - a_1 \int t^{-s} d\nu \right| \\ \leq \frac{|C|}{L} \int t^{1-s} d\nu \rightarrow 0. \end{aligned} \quad (\text{B3})$$

The moments used are finite: on $(0, T]$, $t^{-s} \leq T t^{-(s+1)}$ and $t^{1-s} \leq T^2 t^{-(s+1)}$, and $\int t^{-(s+1)} d\nu$ is finite by assumption. Finally, the Mellin transform gives $\int_0^\infty f(u) u^{s-1} du = \Gamma(s) \int t^{-s} d\nu$, so $\int t^{-s} d\nu = M_f$. $\square$ The rest of the formal proof establishes integrability for Fubini and for each moment, and is routine.

Appendix C: One to six, derived

A species' operator is its kinetic operator D_0 with a curvature term inserted. Under the cut in the species'

own operator (K1), the curvature coefficient can be read in two ways, and the two agree for every proper-time profile.

a. The full operator. A constant insertion δ shifts the spectrum by δ , so the heat trace of $D_0 + \delta$ is $e^{-\tau\delta} \Theta(\tau)$. Expanding the product,

$$\tau^{s+1} e^{-\tau\delta} \Theta(\tau) = a_0 + (a_1 - \delta a_0) \tau + O(\tau^2), \quad (\text{C1})$$

with an explicit constant in the remainder. So the shifted operator has $a_1(D_0 + \delta) = a_1(D_0) - \delta a_0(D_0)$: Gilkey's curvature term for a constant insertion, derived rather than imported. By Appendix B, its cutoff trace has coefficient $M_f (a_1 - \delta a_0)$.

b. The kinetic part plus the response. The kinetic part is the cut in D_0 , with coefficient $M_f a_1$. The insertion, read as the first-order response of the cutoff trace,

$$\mathcal{R}(L) = \frac{d}{d\delta} \text{Tr } f\left(\frac{D_0 + \delta}{L}\right) \Big|_{\delta=0} = - \int \frac{t}{L} \Theta\left(\frac{t}{L}\right) d\nu(t), \quad (\text{C2})$$

has coefficient $-a_0 \mathcal{D}/\Gamma(s+1)$ at order L^s , where $\mathcal{D} = - \int_0^\infty f'(u) u^s du$ is the profile's derivative moment. The estimate is the one in Appendix B, one power of t higher. Differentiating under the integral takes uniform bounds, which are routine.

c. The integration by parts. The Mellin transform gives $\mathcal{D} = \Gamma(s+1) \int t^{-s} d\nu$, so $\mathcal{D}/\Gamma(s+1) = M_f$. For proper-time profiles, the integration by parts between the two moments is the Gamma function's recursion, $\Gamma(s+1) = s\Gamma(s)$.

d. One to six. Write $\mathcal{I} = (4\pi)^{-(s+1)} \int \sqrt{g} R$. With the kinetic operator's $a_1 = (N/6)\mathcal{I}$ and an insertion of trace c , so that $\delta a_0 = c\mathcal{I}$, both readings give

$$M_f \frac{N}{6} \mathcal{I} - \delta a_0 M_f = M_f \frac{N - 6c}{6} \mathcal{I}. \quad (\text{C3})$$

The kinetic part and the insertion stand at one to six, with the same factor M_f , for every proper-time profile. An insertion that varies over the background is not derived this way; for it the coefficient is Gilkey's, as imported in Appendix A.

e. Profiles of bounded variation. A profile need not be differentiable; the sharp step is the obvious case. A profile of bounded variation can be written through its derivative measure, $f(u) = m_+((u, \infty)) - m_-((u, \infty))$, with $m_\pm$ carried by $(0, \infty)$ and of finite s -th moment. For each side, the layer-cake formula gives

$$\int u^s dm = s \int_0^\infty m((t, \infty)) t^{s-1} dt, \quad (\text{C4})$$

so $\int u^s d(m_+ - m_-) = s \int_0^\infty f(u) u^{s-1} du$, with no derivative and no boundary term. The insertion's weight relative to the kinetic part is therefore six for every such profile with non-zero moment. What turns that weighting into the induced coefficient is the spectral-action expansion, which the literature states for Laplace-transform and for Schwartz profiles [9–11] and which is taken here as a hypothesis for them, not proved. For the sharp step it holds beyond the leading term only as a Cesàro mean [11].

Appendix D: The counts in closed form

a. Conventions. Fermions are counted as Weyl (two-component) fermions; Dirac and Majorana counts are derived from them, never entered separately. A species' field components N are its covariant components, one per real field, the convention its heat-kernel coefficient is stated in. The horizon count uses a different quantity, each species' coefficient in the horizon's entanglement entropy in units of one real scalar (Table II), which is not a count of physical degrees of freedom. Each gauge vector is counted with its two Faddeev–Popov ghosts, minimally coupled scalars of opposite statistics. The contact term is assigned to the vector, for the gauge field with its ghosts, as Kabat states it [13]; no other species has one. Each of the 12 vectors counts as one abelian gauge field, which is exact at one loop on a purely gravitational background.

b. The closed form. With the Standard Model's gauge group and its ghosts, a content of g generations, h Higgs doublets and n right-handed neutrinos has $15g + n$ Weyl fermions of weight $+1$, $4h$ real scalars of weight $+1$, 12 vectors of weight -2 and 24 ghosts of weight -1 . Its net count is Eq. (8),

$$N_{\text{net}}(g, h, n) = 15g + 4h + n - 48. \quad (\text{D1})$$

Every statement below holds for all non-negative integers; none rests on a search.

c. The floor. With one doublet, $N_{\text{net}} = 15g - 44$, which is positive exactly when $g \geq 3$. The values for zero to six generations are those of Fig. 1.

d. The least content. A count of exactly one needs $15g + 4h = 49$. Reducing modulo four, $-g \equiv 1$, so $g \equiv 3 \pmod{4}$. Then $g = 3$ gives $h = 1$, and $g \geq 7$ overshoots, so three generations and one doublet is the only solution. With a right-handed neutrino in every generation, the condition becomes $(15 + 1)g + 4h = 49$, which has no solution, since the left side is a multiple of four and the right side is not. As a cross-check the repository also searches every content with fewer than 12 generations and fewer than 20 doublets, a range that covers every possible solution; it finds the same.

e. Right-handed neutrinos. Adding n of them to the Standard Model gives a count of $1 + n$, a kinetic total of $-62 - 2n$ and an insertion total of $+63 + 3n$. So, with $N_{\text{kin}} = -62$ and $N_{\text{ins}} = +63$, the threshold is $6(|N_{\text{kin}}| + 2n)/(N_{\text{ins}} + 3n)$ and the margin is $(1 + n)/(N_{\text{ins}} + 3n)$, which increases with n . The sign never changes.

f. The curvature coupling. Give the doublet's 4 real scalars a common coupling ξ , so each has curvature trace ξ and weight $1 - 6\xi$. The Standard Model's count becomes $-3 + 4(1 - 6\xi) = 1 - 24\xi$, positive exactly when $\xi < 1/24$. In general, a content whose other species contribute r and whose k scalars share the coupling is positive exactly when $\xi < (r + k)/(6k)$, which gives $1/8$ and $1/6$ with two and three right-handed neutrinos. A single scalar coupled on its own faces a different bound; it is the common coupling that gives $1/24$. At the conformal

coupling $\xi = 1/6$ the count is -3 , -1 and 0 with none, two and three right-handed neutrinos.

g. Gauge bosons. A content of k_s minimally coupled real scalars, k_W Weyl fermions and v gauge vectors, each with its two ghosts, has $N_{\text{net}} = k_s + k_W - 4v$. With the Standard Model's 4, 45 and 12 this is $+1$, so b further vectors and k further scalars or Weyl fermions give $1 + k - 4b$, which is positive exactly when $k \geq 4b$, since both sides are integers.

h. The horizon identity. Over any content built from the four species, the horizon count, entanglement coefficients plus contact terms, differs from the induced count by

$$(\text{entanglement} + \text{contact}) - N_{\text{net}} = n_{\text{ghost}} - 2n_{\text{vector}}. \quad (\text{D2})$$

So the two agree exactly when every vector carries its two ghosts, which is condition K2. For the Standard Model they read $+73$ and -72 , a net of $+1$. That the horizon count *should* equal the induced one is the proportionality between horizon entropy and the induced coupling [12, 15], which is cited here, not proved.

Appendix E: The magnitude

In four dimensions, Eq. (A5) gives $(16\pi G)^{-1} = M_f N_{\text{net}} L / (192\pi^2)$, so a content of net count N_{net} induces $G = M_P^{-2}$ exactly when

$$\Lambda = M_P \sqrt{\frac{12\pi}{M_f N_{\text{net}}}}. \quad (\text{E1})$$

The normalized moment is $1/\Gamma(s + 1)$ for the sharp step $\theta(1 - u)$, 1 for the heat kernel's profile, and $\Gamma(s/2)/(2\Gamma(s))$ for the Gaussian e^{-u^2} . In four dimensions these are 1, 1 and $\sqrt{\pi}/2 \approx 0.8862$. For the Standard Model this gives 6.140 to 6.522 M_P (Table V). The heat kernel's profile is a proper-time profile, so its row rests on Appendix B; the Gaussian's rests on the imported expansion for Schwartz profiles; and the sharp step's rests on an expansion that holds beyond the leading term only as a Cesàro mean (Appendix C). The integration by parts, and these moments, are also checked numerically in the repository against the closed forms, at $s = 1$ and $s = 2$, with the quadrature's tolerance checked again in a second software environment with independently installed library versions. The magnitude moves with the profile; the sign does not.

Appendix F: The formal statements

Every result in Appendices A–E is proved in Lean 4 against Mathlib, in the repository's Lean library, under the checks described in Sec. X; imported results enter only as hypotheses the theorems take. Table VI lists every result of the main text that cites Lean: its key in the

TABLE VI. The results that cite Lean: each result's key in the repository's CLAIMS.md, the section in which it appears, and the Lean statements it cites.

Result (CLAIMS.md key)	Sec.	Lean statements cited
induced-newton-constant-positive	IV	sm_newton_positive, sm_newton_positive.properTime, sm_newton_positive.monotone, ProperTimeProfile.spectralActionA1, InducesNewton.unique, attractive_iff
headline-count	IV	sm_kinetic, sm_insertion, sm_net
sign-profile-independent	III	inducesNewton.of_spectralAction, inducesNewton.properTime, ProperTimeProfile.normalizedMoment_pos, normalizedMoment_pos.ofMeasure, one_to_six, Profile.weighting_eq_six
horizon-independent-route	V	sm_count, horizon_agrees_iff
species-weight	III	Generated.SpeciesRow.weight_eq_kinetic.add.insertion, netTotal_eq_kinetic.add.insertion, one_to_six, inducesNewton.of_spectralAction
one-to-six	III	one_to_six, ProperTimeProfile.hasDerivAt.F, ProperTimeProfile.derivativeMoment_eq, Profile.derivativeMoment_eq, Profile.insertionMoment_eq, Profile.sharp_f, Profile.sharp_moment
criterion	IV	criterion, sm_threshold, sm_criterion_iff
criterion-margin	IV	sm_margin, Profile.weighting_eq_six, one_to_six
profile-family-weighting	III	Profile.weighting_eq_six, Profile.sharp_f, Profile.sharp_moment
standard-model-count	IV	sm_kinetic, sm_insertion, sm_net
horizon-count	V	sm_count, horizon_agrees_iff
generation-floor	VI	net_smContent, generation_counts, generation_floor, generations_newton
unique-least-content	VI	net_smContent, unique_count_one, no_count_one_with_neutrinos
coupling-bound	VI	sm_coupled_newton, sm_coupled_newton.properTime, sm_higgs_net, gaugeFieldsCounted.branch0
neutrino-margins	VI	margin_two, margin_three, sm_margin, margin_neutrinos_lt
neutrino-sign-robust	VI	net_neutrinos, neutrinos_newton_positive
magnitude-range	VII	cutoff_for_newton, sm_cutoff_for_newton, normalizedMoment_sharp_four, normalizedMoment_heatKernel, normalizedMoment_gaussian_four
gauge-boson-cost	VIII	gauge_vector_cost, net_gaugeContent, smContent_eq_gaugeContent
extension-rule	VIII	net_smExtension, smExtension_pos_iff, extension_newton, sm_plus_one_boson, sm_plus_boson_and_scalars, agree_extensions
conformal-higgs-excluded	VIII	sm_conformal_net, sm_conformal_not_attractive, sm_higgs_net, agree_conformal

repository's CLAIMS.md, where its wording and evidence

are given; the section of this paper in which it appears; and the Lean statements it cites.

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